Fractional Fourier-Enhanced Fusion Network Based on Pareto Optimization for Hyperspectral and LiDAR Data Classification

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***Abstract*— In recent years, the utilization of hyperspectral images (HSIs) and light detection and ranging (LiDAR) for collaborative classification has emerged as a significant research direction in Earth observation tasks, with diverse joint classification algorithms showing promising performance using varying network architectures. However, these methodologies infrequently address the challenge of fusion arising from the substantially larger volume of HSI feature information compared to LiDAR features. Moreover, the effective learning of HSI and LiDAR features while mitigating modality conflicts remains an area that necessitates further investigation. As such, a fractional Fourier-Enhanced fusion network (FrFENet) based on Pareto optimization is proposed for HSI and LiDAR data classification. To address the disparity in information volume between modalities, a weighted fractional Fourier-Enhanced fusion (WFrFEF) module is introduced, which applies a weighted fractional Fourier transform (WFrFT) to HSI features, enhancing their representations and facilitating balanced fusion with LiDAR features. Furthermore, a Pareto-based soft optimization strategy, HLPareto, is designed to balance learning rates across HSI and LiDAR features in a dual-branch network, effectively avoiding optimization conflicts. Additionally, a spatial–spectral integration module (SSIM) and an elevation information enhancement module (EIEM) are developed to improve feature extraction. The SSIM enables effective spatial–spectral fusion by facilitating token-level interactions, while the EIEM enhances elevation feature representation, preserving spatial geometric information in LiDAR data. Extensive experiments and comparative analyses conducted on three widely utilized HSI and LiDAR datasets have shown that the proposed FrFENet exhibits superior classification performance.**

***Index Terms*— Collaborative classification, hyperspectral image (HSI), light detection and ranging (LiDAR), Pareto optimization, weighted fractional Fourier.**

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I. Introduction

HYPERSPECTRAL imaging technology facilitates the comprehensive acquisition of continuous spectral information across the ultraviolet, visible, near-infrared, and mid-infrared regions [1]. The exceptionally high spectral resolution of this technology enables pixel representations that more accurately reflect the physical properties of the observed target [2]. Notably, hyperspectral images (HSIs) encompass not only 1-D features characterizing the spectral response but also 2-D geometric information depicting the target distribution [3], [4]. Consequently, hyperspectral imaging finds extensive application in diverse fields, including pollution monitoring [5], urban planning [6], geological exploration [7], and precision agriculture [8].

HSI classification (HSIC) is a crucial technique for interpreting HSIs, which utilizes a multidimensional feature space and assigns a corresponding class label to each pixel [9]. To classify HSIs, numerous classification methods have been gradually developed [10], [11], [12], [13], [14]. However, due to the influence of natural environmental conditions such as salinity and moisture, some vegetation types may exhibit multiecological characteristics, which are characterized by the phenomenon of the same species with different spectra in HSIs [15]. Furthermore, land covers that are similar but belong to different categories often exhibit analogous spectral signatures in HSIs, further exacerbating the challenges of HSIC [16]. Fortunately, different land-cover types exhibit variations in geometric morphological features and vertical elevation distribution characteristics, providing novel opportunities for fine-scale land-cover classification.

Based on the above, researchers hope to supplement hyperspectral data by introducing other data. Light detection and ranging (LiDAR) [17] offers the advantage of not being susceptible to atmospheric interference, such as clouds or fog, compared to hyperspectral imaging. LiDAR data provide reliable ground elevation information in the form of numerical arrays, which can complement hyperspectral data at the

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elevation level for more precise identification of specific land-cover classes [18]. Consequently, researchers are increasingly focusing on the joint classification of HSI and LiDAR data.

To leverage the advantages of HSI and LiDAR data in classification tasks, some traditional machine learning (ML) methods have been developed. In article [19], HSI and LiDAR features are classified independently utilizing support vector machine (SVM) classifiers, with subsequent decision fusion implemented through the application of naive Bayes methodology. Chen et al. [20] proposed a multiprobability decision fusion method for different features of HSIs and LiDAR. To fully exploit the discriminative information in HSI and LiDAR data, morphological profiles (MPs) [21], attribute profiles (APs) [22], and extinction profiles (EPs) [23] are widely employed during the feature extraction process. These features are subsequently concatenated and fed into carefully designed classifiers. However, the straightforward concatenation of these high-dimensional features inevitably results in the Hughes phenomenon [24]. To address this issue, several dimensionality reduction methods [25], [26], [27] have been proposed to reduce feature redundancy. Although the aforementioned ML-based methods have shown promising results, these traditional approaches rely heavily on manual feature engineering, have limited capacity to model nonlinear relationships, and are less effective in handling large-scale complex HSIs.

With the rapid development of deep learning (DL), many novel DL methods have been proposed for HSI and LiDAR data classification. These methods are attributed to their superior capability in extracting high-level, multilayered, and abstract features from input data, as well as in managing high-dimensional multimodal features [28], [29]. Hang et al. [30] employed a convolutional neural network (CNN) with two parameter-sharing branches to extract features, thereby facilitating the learning process between the two modalities. Zhao et al. [31] employed fractional Gabor convolutional (FGC) layers to extract multiscale and multidirectional features from HSIs and LiDAR for fusion classification. Zhang et al. [32] proposed the interleaving perception CNN (IP-CNN), which mitigated the complexity of fusing heterogeneous features from HSIs and LiDAR. Nevertheless, CNNs are inherently inclined toward local feature perception, making it challenging to directly capture long-range global contextual information [33].

In contrast, Transformers, leveraging attention mechanisms, excel in capturing dependencies across all positions in the input sequence [34], [35]. As a result, a set number of CNN–Transformer-based approaches have been proposed in recent years for the joint classification of HSI and LiDAR data. Recently, Roy et al. [36] proposed a multimodal fusion Transformer (MFT) that extracts features from HSI data and integrates them with classification (CLS) tokens derived from LiDAR data, significantly improving joint classification performance. Chang et al. [37] employed a hierarchical approach to achieve multimodal local–global feature extraction, further enhancing the classification accuracy.

Intriguingly, to transcend the constraints posed by spatial and channel dimensions, recent advancements have progressively incorporated frequency-domain information into the realm of multimodal learning. Ni et al. [38] innovatively utilized fast Fourier transform (FFT)-based techniques to adaptively acquire feature representations from HSI and LiDAR data. Furthermore, the fractional Fourier transform (FrFT) serves as an extension of the traditional Fourier transform (FT), allowing for the rotation of signals at any angle between 0 and $\pi/2$ within the time–frequency plane [39]. This functionality provides a more nuanced analysis of signals that exist in the transitional space between the time and frequency domains. Recently, Zhao et al. [40] delved into the exploration of a fractional Fourier image Transformer (FrIT), which effectively extracts global contextual information, thereby minimizing the redundancy of information as it traverses from shallow to deep layers. Despite significant progress in feature extraction methods, two critical issues remain unresolved. First, effectively leveraging patch-level features that share the same land-cover label as the central pixel remains a pressing challenge. Simultaneously, achieving minimal interference from irrelevant pixels during the extraction of high-level features necessitates equal attention. Second, due to the imbalance in information between HSIs and LiDAR, the model optimization process typically exhibits a bias toward the HSI modality, which is more easily fit, thereby leading to inadequate learning of LiDAR features. Consequently, balancing the learning rates of HSI and LiDAR features in neural networks, fully exploiting the potential of each modality, and enhancing the quality of feature mapping are essential for further improving joint classification performance.

In addition, after extracting deep-level multimodal features, it is necessary to fuse the features of different modalities to generate unified representational features, which are subsequently mapped to the classifiers. To utilize the complementary advantages of HSI and LiDAR data and to explore the interdependence between the two modalities more comprehensively, researchers have proposed various interaction fusion methods. For instance, Dong et al. [41] employed an attention-based cross-guidance methodology to produce more compact features from the fusion of HSIs and LiDAR. Furthermore, other interaction fusion networks [42], [43], [44], [45] have also shown good performance. However, these approaches typically rely on the attention mechanism to establish connections between different modalities to facilitate the exchange and integration of information. Nevertheless, a significant issue that warrants consideration is that the HSI contains substantially more information than LiDAR, and this disparity in the quantity of information between modalities can readily lead to the overshadowing of LiDAR information during the fusion stage, thereby potentially diminishing the performance of the joint classification task.

To address the aforementioned issues, a fractional Fourier-Enhanced fusion network (FrFENet) based on Pareto optimization is proposed for HSI and LiDAR data classification. Specifically, to mitigate significant discrepancies in information volume between modalities, we design a weighted fractional Fourier-Enhanced fusion (WFrFEF) module. Benefiting from the decorrelation property of the FrFT, this module applies a weighted fractional Fourier transform

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(WFrFT) to HSI features, thereby enhancing feature representations and generating balanced fused features in combination with LiDAR features. Subsequently, to balance the learning rates of HSI and LiDAR features in a dual-branch neural network, a Pareto soft optimization strategy tailored for joint HSI and LiDAR classification, termed HLPareto, is introduced. This strategy effectively prevents optimization conflicts between the two modalities during the pursuit of a unified learning objective. Finally, to effectively utilize patch features sharing the same land-cover label as the central pixel, we develop a spatial–spectral integration module (SSIM) and an elevation information enhancement module (EIEM) for the HSI and LiDAR feature extraction stages. The SSIM facilitates spatial–spectral feature fusion through token-level information interaction, effectively reducing interference from irrelevant pixels. Meanwhile, the EIEM enhances the representation of elevation information while preserving the rich spatial geometric features inherent in LiDAR data.

In summary, the main contributions of this article can be summarized as follows.

1) To mitigate the heterogeneity-induced information asymmetry in cross-modal feature fusion, this article proposes the WFrFEF. The core innovation of this module lies in its frequency-domain feature correction mechanism, which effectively alleviates the dominance of high-information HSI features while preventing the suppression of low-information LiDAR representations. Specifically, the module first applies the WFrFT to enhance hyperspectral features in the frequency domain, followed by the integration of LiDAR features to construct a balanced fusion representation.
2) Considering the competitive imbalance in deep models when learning HSI and LiDAR features, this article proposes a Pareto-based soft optimization strategy (HLPareto) to balance the network's learning rates for both modalities. This strategy effectively mitigates optimization conflicts between the two modalities for a shared learning objective through gradient-based optimization, alleviating imbalanced training within the model.
3) To reduce the interference of irrelevant pixels, the SSIM and the EIEM are developed to explore patch features sharing the same land-cover label as the central pixel. In the HSI branch, the SSIM fuses features by capturing semantic similarities and differences between spatial and spectral features at the token level. In the LiDAR branch, the EIEM enhances the representation of elevation information while preserving the rich spatial geometric features of LiDAR data.

The remainder of this article is organized as follows. In Section II, the proposed FrFENet is presented in detail. Section III reports and analyzes experimental results. Finally, conclusions are outlined in Section IV.

II. Methodology

Fig. 1 presents an overview of the FrFENet. During forward propagation, the network architecture consists of two main components: dual-branch feature extraction and information fusion. Specifically, in the HSI feature extraction branch, a dual-branch design is employed to extract spectral and spatial features, followed by SSIM to facilitate efficient spatial–spectral feature fusion. For LiDAR feature extraction, a 2-D residual network extracts elevation features and enhances their representation with EIEM. Subsequently, a WFrFT is applied during the information fusion stage to generate relatively balanced multimodal feature representations. Finally, to effectively address optimization conflicts between the dual-branch modalities while pursuing a unified learning objective, an HLPareto is employed for weight updates during backward propagation.

A. HSI Feature Extraction Branch

To extract the rich spectral and spatial information in an HSI, a dual-pathway network is employed for spatial–spectral feature extraction. One pathway employs a 1-D residual block (ResBlock) to extract spectral features, capitalizing on the excellent learning capability of 1-D convolution for high-dimensional sequences. Concurrently, the other pathway utilizes Transformer blocks to capture spatial features, leveraging the ability to capture global contextual information.

Assume that the HSI training samples can be denoted as $\boldsymbol{D}^h = \{\mathbf{x}_i^h\}_i^N \in \mathbb{R}^{p\times p\times d}$, the corresponding labels are denoted as $\mathbf{Y} = \{\mathbf{y}_i\}_i^N \in \mathbb{R}^{1\times C}$, where p is the patch size, d is the spectral dimension, N is the total number of samples, and C is the number of classes. To extract spectral features, the central pixel $\mathbf{x}_i^{\text{spec}} \in \mathbb{R}^{1\times d}$ of the training patch is used as input. As shown in Fig. 2(a), a 1-D ResBlock is employed for HSI spectral feature extraction. The 1-D ResBlock primarily consists of a 1-D convolutional layer, a batch normalization (BN) layer, and a Leaky ReLU layer. Subsequently, an AvgPool1D layer and a linear layer are introduced to further map the features into a deeper latent space. The output of the overall ResBlock can then be expressed as

$$\boldsymbol{f}_i^{\text{spec}} = \varphi_{\text{spec}}\left(\mathbf{x}_i^{\text{spec}}\right), \quad \boldsymbol{f}_i^{\text{spec}} \in \mathbb{R}^{1\times B} \tag{1}$$

where the B is the dimension of latent space, φ_{spec} represents the overall 1-D ResBlock, and $\boldsymbol{f}_i^{\text{spec}}$ represents the final spectral feature of the HSI.

In terms of spatial pathways, the Vision Transformer (ViT) is employed to extract spatial-domain features. As shown in Fig. 1, HSI patches are first divided into smaller patches through patch embedding, with a learnable class token and position embedding added. Subsequently, the output of the patch embedding is fed into the Transformer block, which primarily comprises normalization layers, a multihead attention layer (with 4 heads), and a multilayer perceptron (MLP) layer. To extract deeper features, 4 Transformer blocks are stacked to produce the final output

$$\boldsymbol{f}_i^{\text{spa}} = \varphi_{\text{spa}}\left(\mathbf{x}_i^h\right), \quad \boldsymbol{f}_i^{\text{spa}} \in \mathbb{R}^{n\times B} \tag{2}$$

where φ_{spa} represents all Transformer blocks, the token count n equals $(p/2)^2 + 1$, and $\boldsymbol{f}_i^{\text{spa}}$ denotes the spatial feature map of the HSI.

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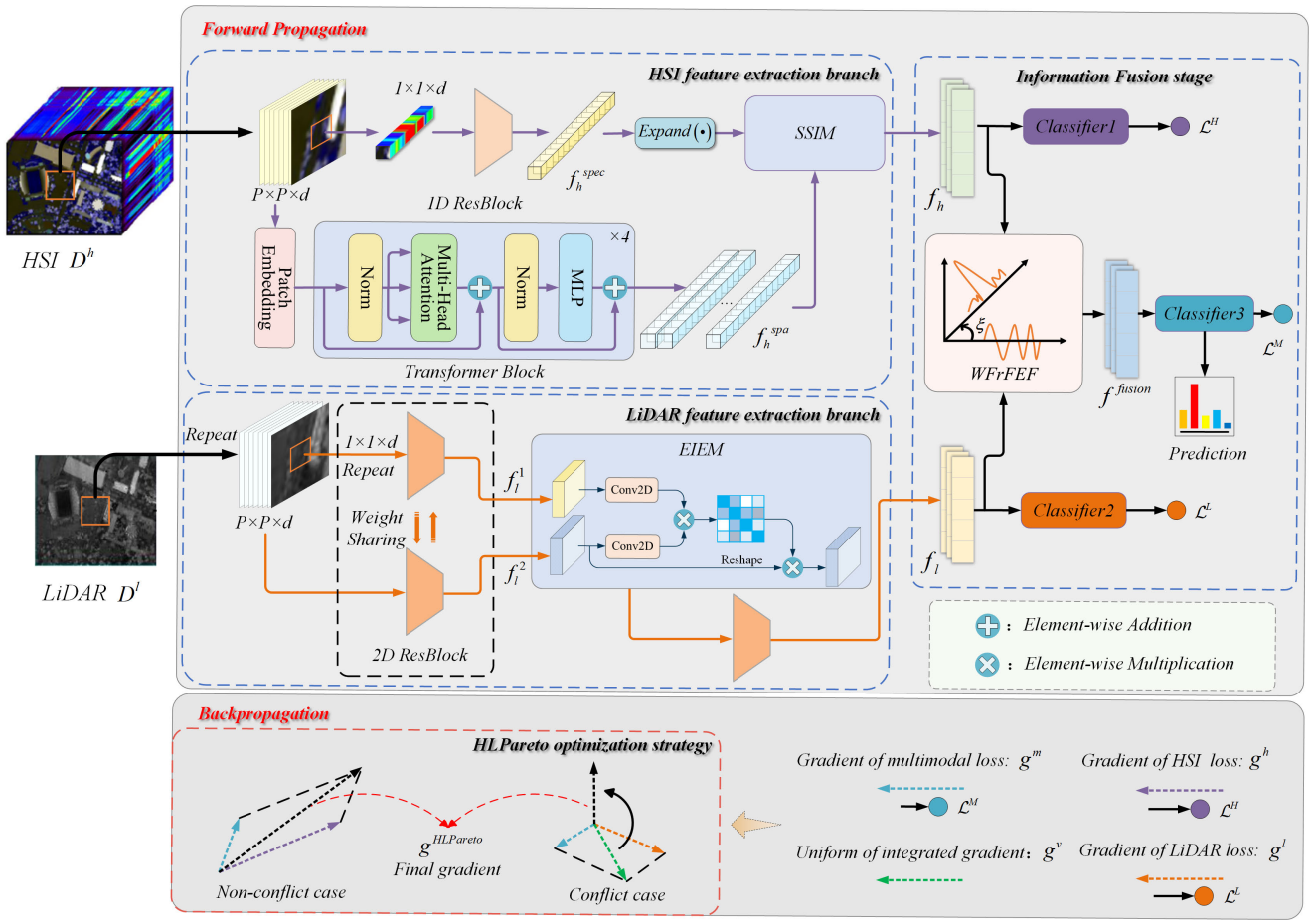


Fig. 1. Overall pipeline of the proposed FrFENet. During the forward propagation process, the designed SSIM facilitates the fusion of spatial–spectral features. For LiDAR feature extraction, the EIEM enhances the representation of elevation features. Subsequently, the WFrFEF is introduced to fuse the two modal features and generate a relatively balanced feature representation. During backward propagation, to balance the learning rates of HSI and LiDAR features in the dual-branch neural network, the HLPareto is employed for gradient backpropagation.

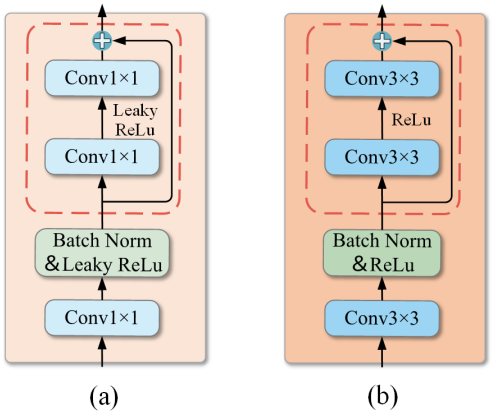


Fig. 2. Details of ResBlock. (a) 1-D ResBlock. (b) 2-D ResBlock.

B. LiDAR Feature Extraction Branch

The LiDAR data provides spatial geometric information and elevation details of surface cover, offering significant potential for feature representation. Given the superior performance of 2-D CNNs in extracting 2-D spatial features, this method employs a 2-D ResBlock for LiDAR feature extraction. To enable the model to extract features more comprehensively, the LiDAR data is first replicated to match the channel number of the HSI, resulting in the training sample $\boldsymbol{D}^l = \{\mathbf{x}_i^l\}_i^N \in \mathbb{R}^{p\times p\times d}$. Subsequently, $\mathbf{x}_i^l$ is input into the designed 2-D ResBlock to extract high-level semantic features. As illustrated in Fig. 2(b), the 2-D ResBlock comprises a convolutional layer, a BN layer, and a ReLU activation function. The output is expressed as

$$\boldsymbol{f}_i^{l,1} = \Phi^1\left(\mathbf{x}_i^l\right), \quad \boldsymbol{f}_i^{l,1} \in \mathbb{R}^{p\times p\times B} \tag{3}$$

where Φ^1 represents 2-D ResBlock and $\boldsymbol{f}_i^{l,1}$ denotes the shallow feature of the LiDAR.

To further extract elevation information, the central pixel of the input patch is fed into another network branch. Specifically, the central pixel is extracted from the sample and expanded by replication to a size of $p \times p$, resulting in $\mathbf{x}_i^{\text{ele}} \in \mathbb{R}^{p\times p\times d}$. Subsequently, $\mathbf{x}_i^{\text{ele}}$ is fed into a network branch with shared parameters for feature extraction, yielding the elevation feature representation

$$\boldsymbol{f}_i^{l,2} = \Phi_{\text{shared}}^1\left(\mathbf{x}_i^{\text{ele}}\right), \quad \boldsymbol{f}_i^{l,2} \in \mathbb{R}^{p\times p\times B} \tag{4}$$

where Φ_{shared}^1 represents the 2-D ResBlock with shared parameters and $\boldsymbol{f}_i^{l,2}$ denotes the elevation feature of the LiDAR.

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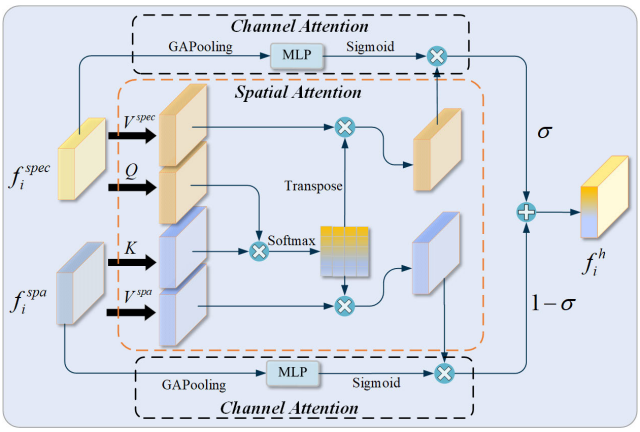


Fig. 3. Details of SSIM.

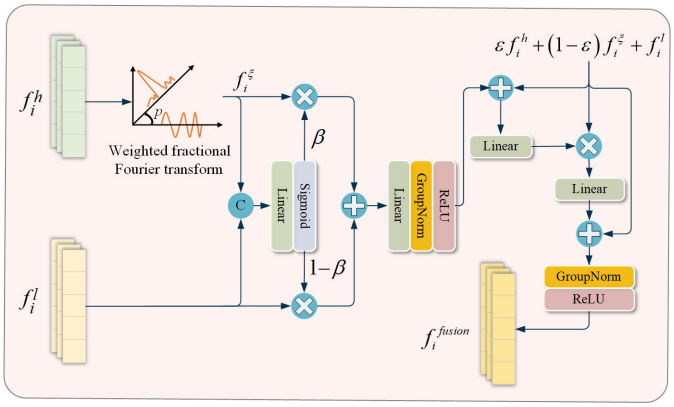


Fig. 4. Details of the WFrFEF module.

C. SSIM and EIEM Modules

HSI and LiDAR data are commonly input into deep models in the form of patches. To effectively utilize patch-level features sharing the same land-cover label as the central pixel while minimizing interference from irrelevant pixels, as illustrated in Figs. 3 and 4, the SSIM and the EIEM were developed. For HSI features, SSIM enhances feature representation by exchanging information between the query of the central pixel and the patch-level support features. For LiDAR features, EIEM computes the affinity between the central pixel's elevation information and each pixel in the patch-level features, reweighting the features to integrate the underlying features of each pixel. The detailed descriptions of these modules are provided below.

1) Spatial–Spectral Integration Module: First, the spectral feature $\boldsymbol{f}_i^{\text{spec}}$ is expanded to the shape of $\boldsymbol{f}_i^{\text{spa}}$ using a vector expansion operation. Convolutional layers are then applied to generate V^{spec} and Q, where $V^{\text{spec}}, Q \in \mathbb{R}^{n\times B}$. Convolutional layers are also applied to the spatial feature to generate V^{spa} and K, where $V^{\text{spa}}, K \in \mathbb{R}^{n\times B}$. Next, the matrix multiplication of Q and K is performed to obtain the feature map, and a Softmax layer is applied to generate the spatial attention map, represented as

$$A = \text{Softmax}\left(\exp\left(QK^{\text{T}}\right)\right). \tag{5}$$

Next, the attention map is multiplied with V^{spec} and V^{spa} through matrix multiplication to obtain the spatially restructured feature map

$$\boldsymbol{F}_i^{\text{spec}} = \text{Trans}_{\text{spec}}\left(AV^{\text{spec}}\right), \quad \boldsymbol{F}_i^{\text{spa}} = \text{Trans}_{\text{spa}}\left(AV^{\text{spa}}\right) \tag{6}$$

where $\text{Trans}_{\text{spec}}$ and $\text{Trans}_{\text{spa}}$ denote transformation functions constituted by convolutional layers.

To enhance feature representation, a channel attention module is employed on the features. Initially, a global average pooling (GAPooling) layer aggregates the original features into a single channel. Subsequently, an MLP layer learns the interdependencies among channels, and a Sigmoid layer generates a channel-wise weighting vector, which is then utilized to reweight the feature map. Finally, a residual connection integrates the output with the original input features, and spatial and spectral features can be adaptively weighted to achieve the final integrated features. The specific computational process is as follows:

$$\hat{\boldsymbol{F}}_i^{\text{spec}} = \text{Sigmoid}\left(\text{MLP}\left(\text{Pool}\left(\boldsymbol{F}_i^{\text{spec}}\right)\right)\right) \odot \boldsymbol{F}_i^{\text{spec}} + \boldsymbol{f}_i^{\text{spec}} \tag{7}$$

$$\hat{\boldsymbol{F}}_i^{\text{spa}} = \text{Sigmoid}\left(\text{MLP}\left(\text{Pool}\left(\boldsymbol{F}_i^{\text{spa}}\right)\right)\right) \odot \boldsymbol{F}_i^{\text{spa}} + \boldsymbol{f}_i^{\text{spa}} \tag{8}$$

$$\boldsymbol{f}_i^h = \sigma\hat{\boldsymbol{F}}_i^{\text{spec}} + (1-\sigma)\hat{\boldsymbol{F}}_i^{\text{spa}} \tag{9}$$

where $\odot$ denotes Hadamard product operation, σ is a learnable balance hyperparameter, and $\boldsymbol{f}_i^h \in \mathbb{R}^B$.

2) Elevation Information Enhancement Module: To suppress the interference of irrelevant pixels in LiDAR features while preserving important geometric information, the EIEM is used to integrate pixel-level features. Specifically, the feature similarity between the pure elevation information $\boldsymbol{f}_i^{l,2}$ and the patch-level feature pixels is computed using cosine distance

$$\text{Sim}\left(\boldsymbol{x}_i, \boldsymbol{z}_j\right) = \frac{\phi(\boldsymbol{x}_i)^{\text{T}}\phi\left(\boldsymbol{z}_j\right)}{\|\phi(\boldsymbol{x}_i)\| \cdot \left\|\phi\left(\boldsymbol{z}_j\right)\right\|}. \tag{10}$$

Here, $\boldsymbol{x}_i$ and $\boldsymbol{z}_i$ denote input feature $\boldsymbol{f}_i^{l,2}$ and $\boldsymbol{f}_i^{l,1}$, with spatial position index i and j. The affinity between the current pixel i and other pixels is then computed using the inner product in the normalized feature space. The similarity is activated by ReLU to suppress negative values, and the final LiDAR feature is the weighted representation of the original a and the similarity

$$\hat{\boldsymbol{f}}_l = \frac{1}{S(\boldsymbol{x}_i)} \sum_{\forall j} \text{ReLU}\left(\text{Sim}\left(\boldsymbol{x}_i, \boldsymbol{z}_j\right)\right)\boldsymbol{z}_j \tag{11}$$

where $S(\boldsymbol{x}_i)$ denotes $\sum_{\forall j} \text{Sim}(\boldsymbol{x}_i, \boldsymbol{z}_j)$. To preserve the original spatial geometric features of LiDAR, the output retains a residual connection. Then, deeper features are obtained through the 2-D ResBlock with the same structure as shown in Fig. 2(b), represented as

$$\boldsymbol{f}_i^l = \Phi^2\left(\hat{\boldsymbol{f}}_l\right), \quad \boldsymbol{f}_i^l \in \mathbb{R}^B \tag{12}$$

where Φ^2 represents 2-D ResBlock with GAPooling along the spatial dimensions and $\boldsymbol{f}_i^l$ represents the high-level elevation features of LiDAR.

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D. Information Fusion Stage

Previous studies [40] have shown that the FrFT has been successfully applied to HSI processing with promising results. As a variant of the FrFT, the WFrFT [46] exhibits a more uniform energy distribution, spreading more widely across the time–frequency plane. Leveraging this property, applying WFrFT to HSI features can effectively reduce redundant information. Specifically, the WFrFT of the HSI feature is represented as

$$\boldsymbol{f}_i^{\xi} = \frac{1}{B}\sum_{n=1}^{B} \boldsymbol{x}(n)\Psi(\xi; n, u) \tag{13}$$

where Ψ is the kernel function of the ξth order WFrFT, $\boldsymbol{x}(n)$ is the value corresponding to the sequence index n in the channel dimension of $\boldsymbol{f}_i^h$, and u represent the fractional domain transformation plane, respectively. The definition of the kernel function is as follows:

$$\begin{aligned}\Psi(\xi; n, u) = \omega_0(\xi)\delta(n-u) + \left(\frac{1}{\sqrt{2\pi}}\right)\omega_1(\xi)\exp(-jnu) \\ + \omega_2(\xi)\delta(n+u) + \left(\frac{1}{\sqrt{2\pi}}\right)\omega_3(\xi)\exp(jnu).\end{aligned} \tag{14}$$

Here, $\omega_i(\xi)$, $i \in (0, 1, 2, 3)$ is the weighting coefficient of the WFrFT, defined as follows:

$$\omega_l(\xi) = \frac{1}{4}\sum_{i=0}^{3}\exp\left[-\frac{j2\pi i(\xi - l)}{4}\right]. \tag{15}$$

In the proposed FrFENet, ξ is adaptively learned. Following the extraction of $\boldsymbol{f}_i^{\xi}$ from the HSI feature through the WFrFT, the initial procedure involves the concatenation of $\boldsymbol{f}_i^{\xi}$ and $\boldsymbol{f}_i^l$, which are subsequently processed through a linear layer followed by a Sigmoid activation function to derive the dynamic weight, denoted as β. Subsequently, these two features undergo reweighting using this dynamic weight and are then subjected to a layer of nonlinear mapping to obtain the preliminary fused feature

$$\hat{\boldsymbol{f}}_i^{\text{fus}} = \text{GN}\Big(\text{ReLU}\Big(W_1\Big(\beta \otimes \boldsymbol{f}_i^{\xi} + (1-\beta)\otimes \boldsymbol{f}_i^l\Big)\Big)\Big) \tag{16}$$

where W_1 denotes the learnable weights and GN refers to Group Normalization.

To preserve the original information of HSI input features and enhance the stability of feature propagation, fusion information $\tilde{\boldsymbol{f}}_i^{\text{fusion}}$, which encapsulates the raw high-level information, is obtained through a weighted summation process. Here, $\tilde{\boldsymbol{f}}_i^{\text{fusion}}$ represents $\varepsilon \boldsymbol{f}_i^h + (1-\varepsilon)\boldsymbol{f}_i^{\xi} + \boldsymbol{f}_i^l$ and ε is a learnable weighting coefficient. Subsequently, the correlation features are derived by multiplying the two fused features

$$\boldsymbol{f}_i^r = W_2\big(\hat{\boldsymbol{f}}_i^{\text{fusion}} + \tilde{\boldsymbol{f}}_i^{\text{fusion}}\big) \otimes \tilde{\boldsymbol{f}}_i^{\text{fusion}} \tag{17}$$

where W_2 denotes the parameters of linear layers. Following this, a residual connection is applied to obtain the final fused features:

$$\boldsymbol{f}_i^{\text{fusion}} = \text{GN}\big(\text{ReLU}\big(W_3\big(\boldsymbol{f}_i^r\big) + \tilde{\boldsymbol{f}}_i^{\text{fusion}}\big)\big). \tag{18}$$

Here, W_3 denotes the parameters of linear layers, which are employed to adjust the dimensionality through linear transformations. This module enhances feature representation by applying a WFrFT to HSI features, constructing a multimodal representation with spatial–spectral coupling characteristics in conjunction with LiDAR features, thereby achieving a more uniform information distribution.

Algorithm 1 FrFENet

Input: Training dataset D^h, D^l, iteration number T, initialized unimodal encoder parameters θ_k for $k = h, l$,
Output: Optimized model parameters θ_k

```
/* Start Training */
for t = 0, 1, ..., T − 1 do
    Sample a mini-batch S from D^h, D^l;
    Feed-forward the data S to the model;
    Backpropagate to compute the gradients;
    for k = h, l do
        Obtain g_S^m and g_S^u for kth unimodal encoder;
        Calculate cos β: β is angle between g_S^m and g_S^u;
        Solve problem of Equation (22), obtain η^m, η^u;
        if ‖η^m g_S^m + η^u g_S^u‖ = 0 then
            Satisfies the Pareto-stationary condition;
        end if
        if cos β ≥ 0 then
            2η^m = 2η^u = 1;
        end if
        Integrate the gradient g_S^HLPareto and update θ_k.
    end for
end for
/* Complete Training */
Save the Network Model Θ
Test the classification accuracy on the testing set D_test
```

E. Pareto Optimization and Backpropagation

Notably, the disparate feature-fitting capabilities of HSIs and LiDAR in the network can lead to imbalanced learning, potentially leading to inadequate utilization of one modality. To address this, additional classifiers are incorporated into both feature extraction branches to enhance single-modal learning. However, optimizing for a shared objective during backpropagation can cause optimization conflicts [47]. To mitigate optimization conflicts between HSI or LiDAR feature branch classifiers and the fused feature classifier, it is crucial to reconcile the gradients between them, allowing the gradients from individual modalities to facilitate multimodal learning in an auxiliary manner. The Pareto principle [48] seeks gradient directions that benefit both objectives to enhance model convergence. Hence, a Pareto-based soft optimization strategy, termed HLPareto, is devised for HSI and LiDAR feature learning. This strategy not only adjusts gradient directions in a gentle, rather than rigid, manner but also regulates gradient magnitudes based on intensity, thereby effectively mitigating optimization conflicts between unimodal and multimodal objectives.

To achieve the aforementioned objectives, classifiers are designed and employed to compute classification losses.

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$\mathcal{L}_{\text{Cls}}$ represents the cross-entropy loss of the classifier, and its computation is given by

$$\mathcal{L}_{\text{Cls}} = \frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} -\boldsymbol{y}_i^c \log \boldsymbol{p}_i^c \tag{19}$$

where p_i denotes the probabilistic output of the classification head. First, three classifiers C_H, C_L, and C_M are designed for the HSI branch, LiDAR branch, and fused features, respectively. The corresponding losses $\mathcal{L}^H$, $\mathcal{L}^L$, and $\mathcal{L}^M$ are computed using the cross-entropy function, and the total loss function is given by

$$\mathcal{L} = \mathcal{L}^M + \mathcal{L}^H + \mathcal{L}^L \tag{20}$$

where $\mathcal{L}$ is total loss. In SGD optimization, for any loss $\mathcal{L}^i$, $i \in (H, L, M)$, the gradient of parameter θ for modality k at the tth mini-batch S is expressed as

$$g_S^i\left(\theta^k(t)\right) = \frac{1}{|S|}\sum_{j}^{|S|} \nabla_{\theta^k(t)} \mathcal{L}^i\left(\boldsymbol{p}_j, \boldsymbol{y}_j\right). \tag{21}$$

To simplify computations, the gradient $\{g_S^i(\theta^k(t))\}_{i\in\{m,u\}}$ for modal k is denoted as $\{g_S^i\}_{i\in\{m,u\}}$. In the context of Pareto's principle, when aiming to balance multiple objective tasks, during each iteration of the model, the gradients corresponding to different objectives are reweighted and combined. Ultimately, the model converges to a tradeoff state along the gradient direction optimized by Pareto, which corresponds to the Pareto-optimality. Specifically, the formula for calculating the Pareto-optimality solution can be expressed as

$$\begin{aligned} &\min_{\eta^m,\eta^u\in} \left\|\eta^m g_S^m + \eta^u g_S^u\right\|^2 \\ &\text{s.t. } \eta^m, \eta^u \geq 0, \eta^m + \eta^u = 1 \end{aligned} \tag{22}$$

where $\|\cdot\|$ denotes the L_2-norm, and the specific computation process is as follows:

$$g_S^{\text{Pareto}}\left(\theta^k(t)\right) = 2\eta^m g_S^m + 2\eta^u g_S^u. \tag{23}$$

However, the direct application of the Pareto algorithm to multimodal models fails to achieve the desired performance. This can be attributed to its implicit impact on the noise intensity of SGD [49]. Furthermore, the rigid alteration of gradient directions adversely affects model performance, particularly as the fitting capacity for HSIs significantly surpasses that of LiDAR. Inspired by paper [50], a more flexible Pareto optimization approach, referred to as HLPareto, has been developed. Specifically, HLPareto is designed to address two distinct scenarios.

1) Nonconflict Case: First, compute the gradient angle β between g_S^m and g_S^u. When there is no conflict between the gradients, $\cos\beta > 0$. In this case, it is unnecessary to employ the Pareto optimization strategy, meaning the gradient at this point is the unified gradient g_S^v

$$g_S^{\text{HLPareto}}\left(\theta^k(t)\right) = g_S^m + g_S^u. \tag{24}$$

2) Conflict Case: When gradient conflicts arise, that is, when $\cos\beta < 0$, the Pareto coefficients are determined using (22) to resolve η_S^m and η_S^u. To facilitate a smoother adjustment rather than a rigid directional change, the original gradients are incorporated into the Pareto-based solution. Furthermore, to enhance generalization, a scaling factor μ is introduced into the gradient optimization process, yielding the final gradient

$$g_S^{\text{HLPareto}}\left(\theta^k(t)\right) = \mu\left(2\eta_S^m + 1\right)g_S^m + \mu\left(2\eta_S^u + 1\right)g_S^u. \tag{25}$$

Referring to paper [50] and to prevent excessive gradients that may lead to model instability, μ is set within the range of 1–2.

Overall, HLPareto accounts for two scenarios. In the case of gradient conflicts, it adjusts both the direction and intensity of the gradient flexibly, implicitly enhancing the SGD noise to stabilize the model and facilitate comprehensive learning of both HSI and LiDAR features. The complete algorithm of the proposed FrFENet method is presented in Algorithm 1, and its schematic is shown in Fig. 1.

III. Experimental Results and Analysis

In this section, we primarily evaluate and discuss the performance of FrFENet on three widely used benchmark HSI and LiDAR datasets. Additionally, the results of the proposed method are compared and analyzed against eight related algorithms. Moreover, ablation experiments are then conducted to verify the effectiveness of different components. Finally, relevant model parameters are discussed and analyzed.

A. Datasets

1) Houston2013: The Houston2013 dataset, acquired from the University of Houston and its surrounding areas, was provided by the IEEE Geoscience and Remote Sensing Society in 2013 [51]. This dataset comprises HSI and LiDAR images with dimensions of 349 × 1905 pixels, both possessing a spatial resolution of 2.5 m. The HSI encompasses 144 spectral bands spanning a wavelength range of 0.38–1.05 μm, while the LiDAR data consists of a single band. The ground-truth data incorporates 15 categories of land cover. Fig. 5 presents the HSI pseudo-color image, the LiDAR image, and the distribution of samples. Details of the training and testing sets are summarized in Table I.

2) MUUFL: The MUUFL dataset was acquired in July 2010 over the University of Southern Mississippi campus [52]. It comprises HSI and LiDAR images with dimensions of 325 × 220 pixels and a spatial resolution of 1 m. The HSI consists of 72 spectral bands encompassing the wavelength range of 375–1050 nm, with 8 initial and final bands removed due to noise, resulting in a total of 64 bands. The LiDAR data contains 2 channels. The ground truth encompasses 11 categories of land cover. Fig. 6 illustrates the HSI pseudo-color image, the LiDAR image, and the distribution of samples. Table II presents details of the training and testing sets.

3) CL-9: The CL-9 dataset was collected in September 2022 in the Tamarix region of the Yellow River Delta, Shandong Province, China [53]. The hyperspectral data were acquired using the Cubert Utilities FIREFL EYE S185 camera,

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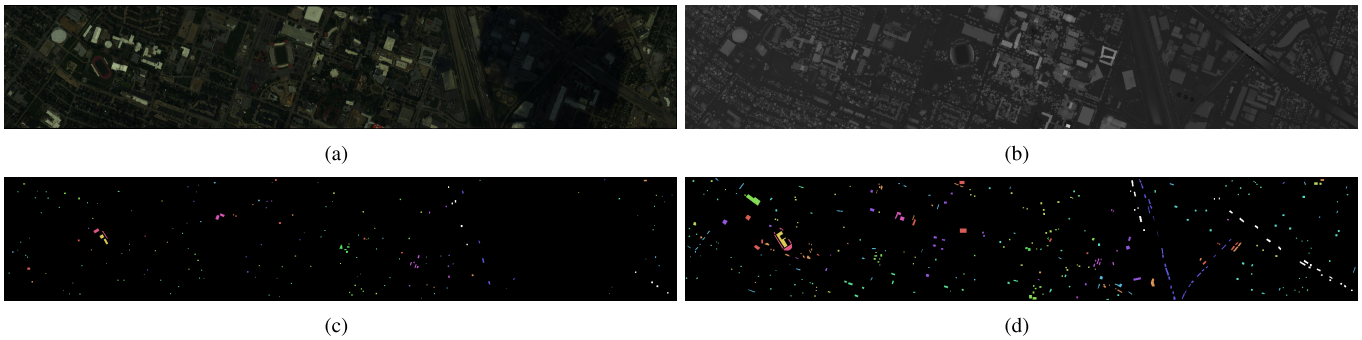


Fig. 5. Overview of the Houston2013 dataset. (a) Hyperspectral pseudo-color imagery. (b) Grayscale image of LiDAR. (c) Training data map. (d) Testing data map.

TABLE I
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE HOUSTON2013 DATASET

No.	Color	Land cover type	Train	Test
1		Healthy Grass	198	1053
2		Stressed Grass	190	1064
3		Synthetic Grass	192	505
4		Trees	188	1056
5		Soil	186	1056
6		Water	182	143
7		Residential	196	1072
8		Commercial	191	1053
9		Road	193	1059
10		Highway	191	1036
11		Railway	181	1054
12		Paking Lot1	192	1041
13		Paking Lot2	184	285
14		Tennis Court	181	247
15		Running Track	187	473
		Total	2832	12197

TABLE II
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE MUUFL DATASET

No.	Color	Land cover type	Train	Test
1		Trees	150	23096
2		Mostly Grass	150	4120
3		Mixed Ground Surface	150	6732
4		Dirt and Sand	150	1676
5		Road	150	6537
6		Water	150	316
7		Buildings Shadow	150	2083
8		Buildings	150	6090
9		Sidewalk	150	1235
10		Yellow Curb	150	33
11		Cloth Panels	150	119
		Total	1650	52037

and the LiDAR data were obtained using the D lidar 2000 camera. Following alignment processing, HSI and LiDAR images with dimensions of 784 × 1204 pixels were obtained. The HSI comprises 126 spectral bands encompassing the wavelength

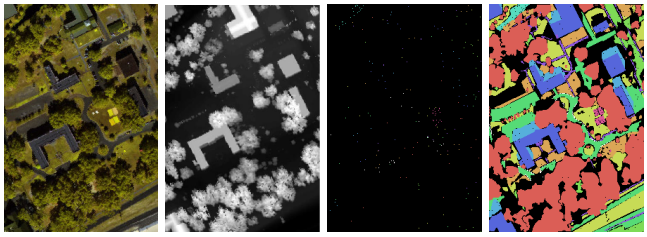


Fig. 6. Overview of the MUUFL dataset. (a) Hyperspectral pseudo-color imagery. (b) Grayscale image of LiDAR. (c) Training data map. (d) Testing data map.

TABLE III
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE CL-9 DATASET

No.	Color	Land cover type	Train	Test
1		Small Tidal Creek	20	4215
2		Phragmites Australis	20	9640
3		Shriveled Spartina Alterniflora	20	52264
4		Abandoned Oil Platform	20	130
5		Tamarix Chinensis (TC)	20	33524
6		Suaeda Glauca (SG)	20	24285
7		Big Tidal Creek	20	47640
8		TC and SG Mixed	20	5428
9		Roads	20	2671
		Total	180	179797

range of 450–950 nm, while the LiDAR data consist of a single channel. The ground truth encompasses 9 categories of land cover. Fig. 7 illustrates the HSI false-color image, the LiDAR image, and the sample distribution. Table III provides details of the training and testing sets.

B. Implementation Details

The proposed FrFENet is implemented in Python 3.7 utilizing the PyTorch framework. All experiments are conducted on an identical hardware platform: GTX-3090 GPU, Intel 4210R CPU, and 30 GB memory. In the network architecture, the spatial size of tokens in the Transformer blocks is set to 2×2, and the embedding dimension is 192. Furthermore, the batch size is set to 32, and the number of epochs is set to 100 during training. For each dataset, 10% of the training set is randomly selected as the validation set. As the proposed algorithm

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TABLE IV
CLASSIFICATION ACCURACIES OF METHODS ON THE HOUSTON2013 DATASET

Class	SVM [10]	CoCNN [30]	S^{2}ENet [55]	GLT-Net [56]	ExViT [57]	Sal^{2}RN [58]	HFSNet [59]	FDNet [38]	**FrFENet**
1	82.34±0.00	82.64±0.22	81.44±0.54	82.81±0.33	81.31±3.91	82.93±0.25	81.04±0.05	**83.04±0.08**	82.85±0.23
2	81.58±0.00	85.06±0.21	84.77±0.24	84.40±0.78	84.66±0.56	**85.11±0.05**	82.52±3.43	83.78±1.39	85.00±0.10
3	**100.0±0.00**	96.40±1.63	98.68±1.02	91.76±4.10	96.59±3.11	97.94±3.94	99.01±1.04	98.02±1.20	98.18±1.71
4	92.42±0.00	93.74±0.70	90.15±1.78	92.16±1.00	94.24±2.94	91.88±0.85	92.88±0.74	94.79±2.22	**98.07±1.53**
5	98.77±0.00	99.98±0.04	97.83±0.16	99.29±0.46	99.32±0.77	**100.0±0.00**	99.98±0.04	99.96±0.05	99.87±0.22
6	95.11±0.00	96.78±3.07	93.71±6.96	96.78±2.07	98.18±2.19	96.22±1.68	89.65±7.22	95.94±0.31	**98.32±1.44**
7	77.15±0.00	79.21±3.90	86.28±2.25	90.21±1.95	88.56±5.60	89.38±3.75	87.84±1.47	88.94±1.63	**93.45±2.01**
8	48.05±0.00	81.39±1.38	79.76±3.76	81.88±4.44	89.16±5.32	79.45±1.03	90.14±5.83	79.57±2.02	**96.07±0.63**
9	78.09±0.00	85.59±5.16	82.91±6.65	89.39±2.04	86.95±2.47	84.17±3.72	85.87±3.33	83.93±1.06	**94.47±1.03**
10	60.43±0.00	59.98±3.81	67.51±0.93	56.62±6.23	63.38±1.85	63.98±7.26	64.52±4.27	65.11±1.03	**69.00±3.13**
11	78.65±0.00	88.67±4.97	89.19±11.6	81.88±7.94	88.12±5.57	86.72±5.53	91.06±3.03	85.50±1.28	**99.43±0.87**
12	78.67±0.00	96.96±2.47	95.06±3.51	91.51±5.00	91.74±2.74	**97.43±2.11**	95.79±3.87	88.78±2.01	96.60±0.74
13	71.93±0.00	90.32±2.46	78.69±7.06	87.30±3.08	87.44±4.20	82.81±4.28	67.46±8.28	**90.88±1.77**	89.33±1.73
14	**100.0±0.00**	97.41±1.91	**100.0±0.00**	93.68±4.33	98.87±±1.89	99.11±0.88	99.60±0.70	99.03±0.68	99.92±0.16
15	98.31±0.00	98.18±2.01	99.92±0.19	93.83±5.33	93.83±8.40	99.96±0.09	97.59±2.49	99.54±0.46	**100.0±0.00**
OA(%)	79.90±0.00	86.80±0.85	86.86±1.74	86.05±1.53	88.04±0.74	87.46±1.46	88.11±0.83	86.97±0.43	**92.32±0.21**
AA(%)	82.77±0.00	88.82±0.70	87.95±1.10	87.57±1.28	89.63±0.51	89.14±1.18	88.60±1.33	89.11±0.41	**93.37±0.22**
Kappa	78.38±0.00	85.72±0.90	85.09±1.39	84.91±1.65	87.04±0.80	86.42±1.57	87.12±0.88	85.92±0.46	**91.68±0.23**

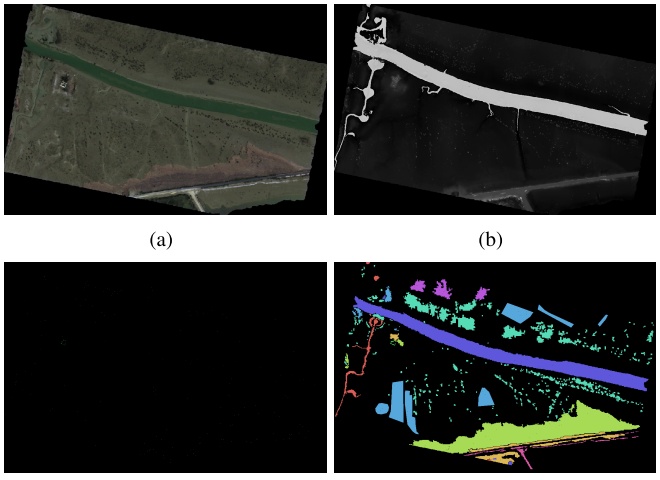


Fig. 7. Overview of the CL-9 dataset. (a) Hyperspectral pseudo-color imagery. (b) Grayscale image of LiDAR. (c) Training data map. (d) Testing data map.

involves Pareto gradient optimization, SGD is employed as the optimizer with a momentum of 0.9, weight decay set to $1e^{-4}$, and the learning rate set to $5e^{-3}$. Four evaluation metrics [54], overall accuracy (OA), class-specific accuracy, average accuracy (AA), and the Kappa coefficient, are utilized to compare the classification performance. Moreover, to mitigate estimation bias resulting from random selection, all experiments were independently conducted ten times.

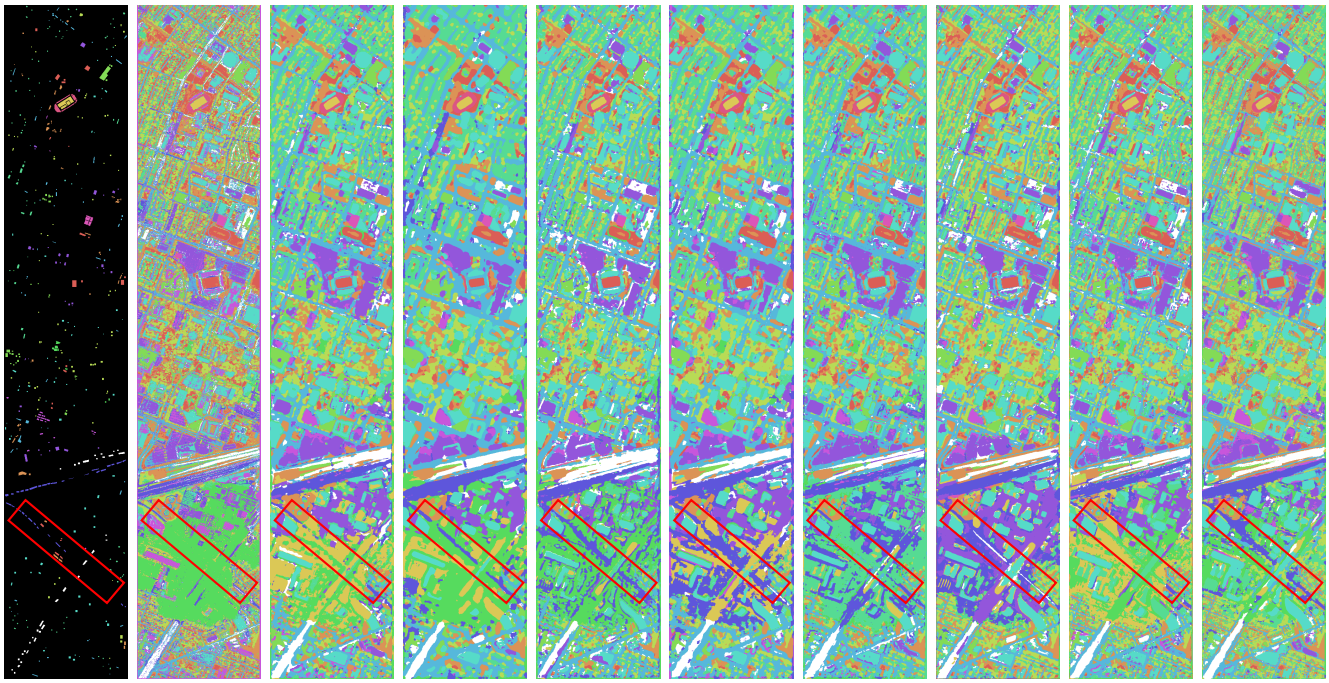
C. Experimental Results and Analysis

To validate the effectiveness of the proposed method FrFENet, eight comparison methods were employed, including a classification method developed specifically for HSIs and seven joint classification methods developed for HSI–LiDAR. The HSI-based method utilizes SVMs [10], while the his–LiDAR joint classification models include a coupled CNN framework (CoCNN) [30], a spatial–spectral cross-modal enhancement network (S^{2}ENet) [55], a global–local Transformer network (GLT-Net) [56], an extended Vision Transformer (ExViT) [57], a spatial–spectral salient reinforcement network (Sal^{2}RN) [58], a height information-guided hierarchical fusion-and-separation network (HFSNet) [59], and a frequency domain-based network (FDNet) [38]. For SVMs, the HSI and LiDAR data are concatenated along the channel dimension before being fed into the model. To ensure the comparability and fairness of the experiments, all models were evaluated on the same hardware platform and configured with the optimal parameters provided in the respective papers.

Tables IV–VI presents the quantitative performance metrics of all methods on three HSI and LiDAR datasets, including class-specific accuracies, OAs, and Kappas, with the best results highlighted in bold. As anticipated, FrFENet exhibited superior performance across three datasets and outperformed the second-highest counterparts by 4.21%, 2.02%, and 1.77% in terms of OA on the Houston2013, MUUFL, and CL-9 datasets, respectively, and achieved the highest performance in terms of AA and Kappa metrics.

For the Houston2013 dataset, the visual and quantitative outcomes of various comparative methods are depicted in Fig. 8 and Table IV, respectively. The dataset's substantial size, dispersed distribution of training samples, and partial cloud cover in certain regions contribute to increased intra-class variability, thereby complicating the differentiation of distinct land-cover classes. From the results, other methods encounter difficulties in distinguishing between "residential" and "commercial" areas. In contrast, the proposed method, FrFENet, exhibits superior classification performance (outperforming the second-best method GLT-Net by 3.24% in OA for "residential" and the second-best method HFSNet by 5.93% in OA for "commercial"). Additionally, as highlighted in the red-boxed sections of Fig. 8, other methods frequently misclassify "railway" as "highway" and fail to effectively leverage the elevation difference features of ground objects, whereas FrFENet exhibits superior classification performance. This observation further underscores the efficacy of the proposed method. Specifically, the SSIM and EIEM enhance feature representation, while the WFrFEF module effectively mitigates

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(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 8. Classification maps of different methods on the Houston2013 dataset. (a) Ground truth. (b) SVM (79.90% OA). (c) CoCNN (86.80% OA). (d) S^2ENet (86.86% OA). (e) GLT-Net (86.05% OA). (f) ExViT (88.04% OA). (g) Sal^{2}RN (87.46% OA). (h) HFSNet (88.11% OA). (i) FDNet (86.97% OA). (j) FrFENet (92.32% OA).

TABLE V

CLASSIFICATION ACCURACIES OF METHODS ON THE MUUFL DATASET

Class	SVM [10]	CoCNN [30]	S^2ENet [55]	GLT-Net [56]	ExViT [57]	Sal^{2}RN [58]	HFSNet [59]	FDNet [38]	FrFENet
1	86.90±0.00	91.18±1.90	91.79±0.75	92.86±1.68	92.78±1.17	93.94±1.42	91.93±1.27	91.82±1.21	**95.55±0.88**
2	71.75±0.00	83.93±4.50	85.29±4.96	89.88±3.72	87.74±2.38	84.31±3.85	88.16±2.75	85.26±4.67	**91.85±1.44**
3	71.84±0.00	83.26±2.67	76.81±6.48	77.72±3.00	85.81±2.08	83.75±3.43	81.22±2.90	78.70±2.56	**85.82±2.31**
4	87.47±0.00	95.27±1.22	89.07±11.9	95.13±1.19	86.52±1.72	94.88±2.77	92.94±1.87	**95.64±2.51**	95.14±1.20
5	90.03±0.00	**93.13±1.44**	89.13±5.05	92.08±1.05	89.84±1.71	91.01±2.31	87.99±2.07	91.10±1.38	92.76±0.90
6	98.42±0.00	**100.0±0.00**	**100.0±0.00**	99.87±0.17	99.94±0.14	**100.0±0.00**	99.94±0.14	99.87±0.28	99.49±0.59
7	88.53±0.00	**97.10±1.62**	95.64±1.79	95.62±0.64	96.03±0.93	94.70±1.43	96.36±1.27	95.13±0.89	95.79±1.50
8	89.31±0.00	95.88±1.48	95.84±2.06	95.91±0.73	95.84±0.49	95.63±0.87	95.45±0.58	95.51±1.09	**96.02±0.41**
9	75.79±0.00	80.62±1.81	86.33±6.85	86.43±2.05	87.17±2.59	87.38±2.19	83.84±2.01	85.12±3.42	**92.29±2.59**
10	**100.0±0.00**	99.39±1.36	93.33±11.6	98.18±1.66	94.55±3.32	96.36±3.95	95.15±5.91	93.33±6.91	**100.0±0.00**
11	99.16±0.00	99.50±0.75	**99.50±0.46**	98.66±1.74	**99.50±0.46**	98.99±0.92	99.33±0.38	99.16±0.59	98.99±0.63
OA(%)	84.35±0.00	90.57±0.58	89.45±1.24	91.02±1.04	91.64±0.59	91.64±0.65	90.24±0.61	90.11±0.52	**93.66±0.34**
AA(%)	87.20±0.00	92.66±0.35	91.15±2.30	92.94±0.31	93.25±0.50	92.81±0.55	92.02±0.73	91.87±1.70	**94.88±0.48**
Kappa	79.63±0.00	87.65±0.72	86.19±1.60	88.20±1.31	89.02±0.75	88.98±0.82	87.17±0.77	87.02±0.71	**91.59±0.44**

TABLE VI

CLASSIFICATION ACCURACIES OF METHODS ON THE CL-9 DATASET

Class	SVM [10]	CoCNN [30]	S^2ENet [55]	GLT-Net [56]	ExViT [57]	Sal^{2}RN [58]	HFSNet [59]	FDNet [38]	FrFENet
1	87.43±0.00	78.50±1.58	86.97±7.26	93.44±4.45	91.85±4.15	**93.49±7.02**	89.96±5.49	91.92±4.19	85.78±6.04
2	74.79±0.00	88.10±0.74	87.66±1.86	88.38±3.83	88.60±1.48	87.47±6.25	90.53±2.56	90.09±3.09	**92.07±2.29**
3	95.99±0.00	96.99±0.34	96.35±3.19	**97.77±1.39**	96.68±0.95	97.69±0.56	96.55±0.87	97.45±0.37	97.71±1.00
4	86.92±0.00	100.0±0.00	99.54±1.03	**100.0±.00**	**100.0±0.00**	**100.0±0.00**	92.15±1.38	100.0±0.00	98.31±1.32
5	47.05±0.00	35.16±5.43	72.99±8.63	69.88±13.3	76.39±8.48	67.18±7.74	60.52±10.2	68.67±6.37	**80.53±2.88**
6	53.22±0.00	**86.04±1.05**	77.86±5.45	85.69±3.45	80.36±4.50	79.72±4.17	79.43±5.73	80.42±1.62	83.17±0.96
7	96.06±0.00	**99.00±0.03**	98.28±0.99	97.37±1.49	97.38±1.23	97.52±2.94	96.75±1.48	97.70±1.92	98.08±0.78
8	44.03±0.00	82.44±2.14	78.70±11.6	67.49±10.4	81.68±9.90	68.54±18.5	63.04±5.00	**85.92±5.02**	85.07±1.01
9	89.89±0.00	96.42±0.71	94.54±1.72	96.93±1.91	97.07±1.94	94.70±2.05	91.88±1.10	**97.22±1.56**	95.72±1.10
OA(%)	78.11±0.00	83.16±1.11	88.76±0.98	89.30±1.94	89.88±1.64	87.96±1.80	86.01±1.30	88.97±1.46	**91.65±0.29**
AA(%)	75.04±0.00	84.74±0.45	88.10±1.43	88.55±1.40	90.00±0.80	87.37±2.37	84.53±1.09	89.93±1.33	**90.71±0.78**
Kappa	72.68±0.00	79.05±1.32	85.89±1.20	86.54±2.42	87.32±2.01	84.89±2.17	82.48±1.58	86.19±1.81	**89.48±0.36**

the information disparity between HSI and LiDAR features during fusion, resulting in high-accuracy classification.

Table IV and Fig. 9 present the quantitative classification results and visual representations of various methods

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TABLE VII
ABLATION STUDY ON THREE DATASETS (%)

Modules				Houston2013			MUUFL			CL-9		
SSIM	EIEM	WFrFEF	HLPareto	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
×	×	×	×	88.38	89.73	87.44	91.25	92.96	88.48	88.82	88.20	85.95
✓	×	×	×	89.77	91.26	88.92	91.80	93.41	89.20	90.35	90.32	87.84
×	✓	×	×	89.02	90.07	88.11	92.28	93.66	89.79	89.16	88.48	86.38
✓	✓	×	×	90.33	91.54	89.53	92.73	93.90	90.37	90.61	89.75	88.19
✓	✓	✓	×	90.40	91.57	89.61	93.03	94.41	90.77	90.85	89.97	88.48
✓	✓	×	✓	91.73	92.94	91.05	93.35	94.67	91.19	91.17	90.27	88.88
✓	✓	✓	✓	**92.32**	**93.37**	**91.68**	**93.66**	**94.88**	**91.59**	**91.65**	**90.71**	**89.48**

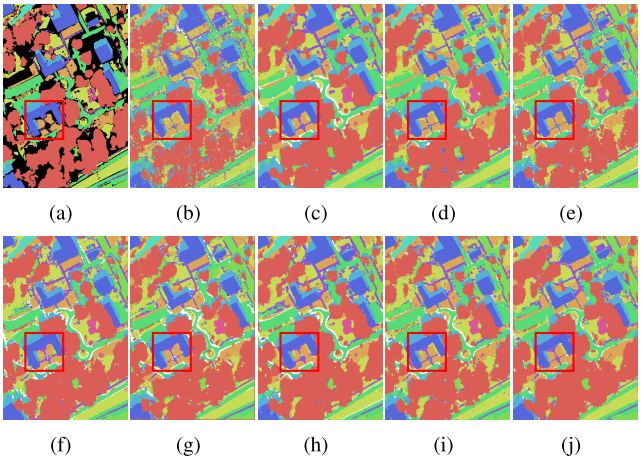


Fig. 9. Classification maps of different methods on the MUUFL dataset. (a) Ground truth. (b) SVM (84.35% OA). (c) CoCNN (90.57% OA). (d) S^{2}ENet (89.45% OA). (e) GLT-Net (91.02% OA). (f) ExViT (91.64% OA). (g) Sal^{2}RN (91.64% OA). (h) HFSNet (90.24% OA). (i) FDNet (90.11% OA). (j) FrFENet (93.66% OA).

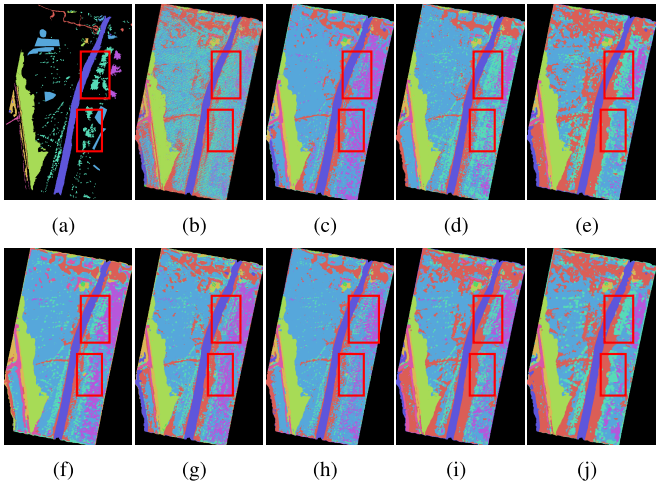


Fig. 10. Classification maps of different methods on the CL-9 dataset. (a) Ground truth. (b) SVM (78.11% OA). (c) CoCNN (83.16% OA). (d) S^{2}ENet (88.76% OA). (e) GLT-Net (89.30% OA). (f) ExViT (89.88% OA). (g) Sal^{2}RN (87.96% OA). (h) HFSNet (86.01% OA). (i) FDNet (88.97% OA). (j) FrFENet (91.65% OA).

on the MUUFL dataset. The quantitative results indicate a prevalent bias among most methods toward fitting the HSI modality, with insufficient utilization of the elevation features derived from LiDAR. For instance, in the categories of "trees" and "mostly grass," the classification performance of other methods is generally suboptimal. However, FrFENet achieves superior results: it outperforms the second-best method, Sal^{2}RN, by 1.61% in OA for the "trees" category and the second-best method, GLT-Net, by 1.97% in OA for the "mostly grass" category. This superiority is attributed to the more balanced treatment of both modality features by the WFrFEF in FrFENet, effectively preventing the high-information HSI from overshadowing the lower-information LiDAR features. As illustrated by the red boxes in Fig. 9, the classification results for "mostly grass" closely align with the ground truth, and the aggregated regions of this category are more complete. Moreover, AA and Kappa coefficients achieved optimal performances of 94.88% and 91.59%, respectively. These results reveal that the proposed HLPareto effectively alleviates the imbalance in model training and enables the model to fully utilize the complementary information from both HSI and LiDAR data sources.

For the CL-9 dataset, the visualization and quantitative results of various comparative methodologies are depicted in Fig. 10 and Table IV, respectively. This wetland dataset is characterized by complex terrain, scattered ground features, and indistinct boundaries, distinguishing it from other datasets. For instance, within the "Tamarix Chinensis (TC)" category, other methods exhibit poor classification performance, frequently misclassifying it as "TC and SG (Suaeda Glauca) Mixed." This misclassification arises due to the spectral similarity among the three types: "TC," "SG," and "TC and SG Mixed." Among these, "TC" exhibits the most prominent elevation and geometric features, followed by "TC and SG Mixed." The proposed SSIM and EIEM can further enhance the feature representation of ground objects, thereby achieving high-precision classification. The quantitative results reveal that despite the limited number of training samples, FrFENet significantly outperforms other methods in the TC category (exceeding the second-best method, GLT-Net, by 10.65% in OA). As illustrated in the red-boxed areas of Fig. 10, FrFENet achieves near-perfect classification of TC and exhibits commendable classification performance for ground features with vague boundaries. This further validates the WFrFEF's role in balancing HSI and LiDAR modalities and the HLPareto strategy's efficacy in maximizing multimodal information utilization.

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D. Effectiveness Analysis of FrFENet

1) Ablation Study: In the proposed FrFENet, three modules—SSIM, EIEM, and WFrFEF—as well as the HLPareto backpropagation optimization strategy were designed. To analyze the contribution of each component to the proposed method, ablation experiments were conducted on three datasets, with the results presented in Table VII. The experiments validated the effectiveness of each component by incrementally adding them to the baseline model, where "✓" indicates addition and "×" indicates no addition. The results suggested that each module improved the classification performance to varying degrees. Specifically, compared to the baseline, SSIM and EIEM improved the OA by 1.95%, 1.48%, and 1.79% on the Houston2013, MUUFL, and CL-9 datasets, respectively. Correspondingly, both AA and Kappa coefficients also increased relatively, indicating that these two modules played a positive role in the HSI and LiDAR feature branches, particularly when processing the features of central and surrounding pixels within patches while reducing the interference from irrelevant pixels.

Furthermore, the WFrFEF module exhibited an improvement in classification performance across all three datasets, highlighting its advantages in balancing modal features and enhancing joint feature representation. After incorporating HLPareto, the OA of all three datasets increased significantly. Notably, there was a 1.40% improvement on the Houston2013 dataset, which features complex and scattered data samples, indicating that this module can effectively mitigate learning conflicts between modalities and fully exploit the potential features of each modality. Ultimately, the experimental results implied that the optimal classification performance was achieved on all three datasets when the four modules were used in combination.

2) Assessment of HLPareto Effectiveness: To further analyze the effectiveness of "HLPareto" in our model, a comparative experiment is conducted across three datasets. Specifically, we selected three modal balancing approaches: 1) "w/o Pareto," where the model was trained in a multitasking manner without employing any Pareto-based gradient optimization strategies; 2) "Pareto," which builds upon multitasking by incorporating traditional Pareto gradient optimization to address directional conflicts between multimodal and unimodal gradients; and 3) "HLPareto," which distinguishes itself from conventional Pareto optimization by not only adjusting directional conflicts but also enhancing gradient magnitudes.

The classification performances (OA) of these three approaches across the three datasets are illustrated in Fig. 11. The figure expresses that "w/o Pareto" marginally outperforms "Pareto" on both the Houston2013 and MUUFL datasets. This observation can be attributed to the fact that rigidly adjusting the gradient directions of multimodal and unimodal objectives potentially diminishes the model's learning capacity. Furthermore, traditional Pareto optimization can affect the noise intensity of SGD, thereby potentially compromising model performance. In contrast, "HLPareto" exhibits superior classification performance compared to the other approaches. This suggests that by softly adjusting gradient directions and magnitudes, "HLPareto" enables the model to more effectively learn from unimodal features and multimodal objectives.

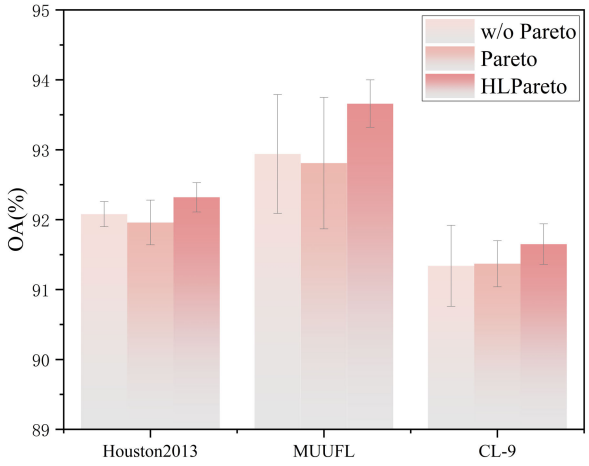


Fig. 11. Effects of HLPareto on the classification performance on three data.

E. Parameter Analysis

1) Impact of Patch Size p: This section examines the impact of the configuration of two parameters in the FrFENet framework on its model performance. Given that the patch size influences the spatial morphology and channel information content of the input pixels from HSI and LiDAR data, we initially investigate the influence of various patch sizes p on the OA of FrFENet's classification performance. Fig. 12 illustrates the relationship between OA and different p across the Houston2013, MUUFL, and CL-9 datasets. During our experiments, while maintaining all other parameters of the FrFENet constant, we adjusted p from the candidate set $\{7 \times 7,\ 9 \times 9,\ 11 \times 11,\ 13 \times 13,\ 15 \times 15,\ 17 \times 17\}$ for experimentation. As depicted in the figure, the OA values for all three datasets exhibit an initial increase followed by a decrease, reaching a maximum at $p = 13$. As p increases from 7 to 13, the OA value progressively rises, indicating that larger patches contribute to uncovering spatial information, thereby enhancing classification accuracy. However, further augmentation in patch size leads to a plateau or even a decline in OA values, potentially due to the introduction of more noisy pixels within larger patches, which compromises the purity of features and consequently affects model performance.

2) Influence Analysis of the Augmentation Factor μ: Since augmentation factor μ controls the intensity of adjustment applied by HLPareto to gradient magnitudes, thereby indirectly influencing the noise intensity of SGD and the generalization capability of the model. Therefore, the influence of various ranges of the augmentation factor μ on the classification performance of three datasets was analyzed.

To ascertain an appropriate setting for μ, we adjusted its value while maintaining other parameters of the FrFENet constant, experimenting with candidates from the set $\{1.0, 1.1, \ldots, 1.9\}$. Fig. 13 displays the trend of OA for this parameter across the three datasets. Overall, the three datasets exhibited low sensitivity to variations in μ, demonstrating

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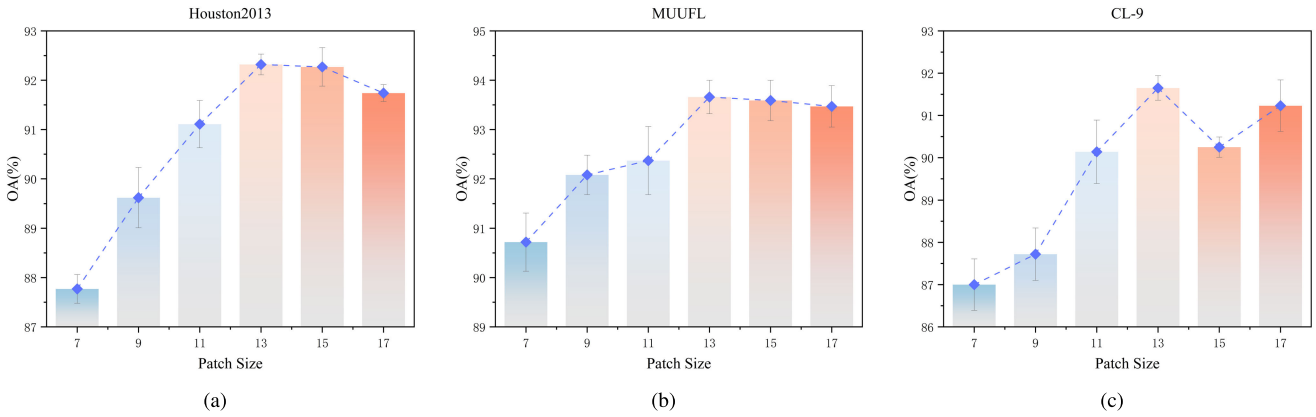


Fig. 12. OA versus patch size on three datasets. (a) Houston2013 dataset. (b) MUUFL dataset. (c) CL-9 dataset.

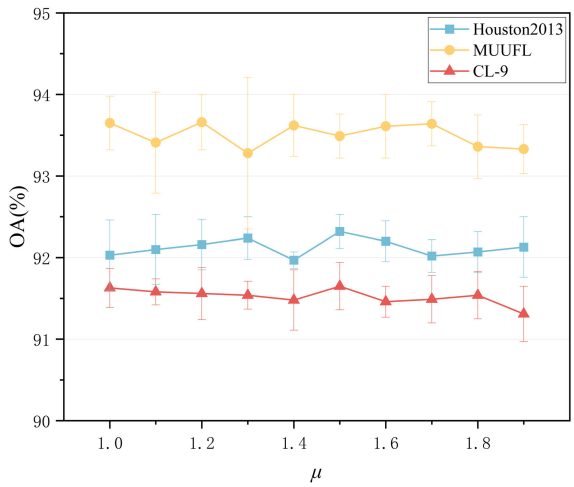


Fig. 13. Effects of μ on the classification performance on three data.

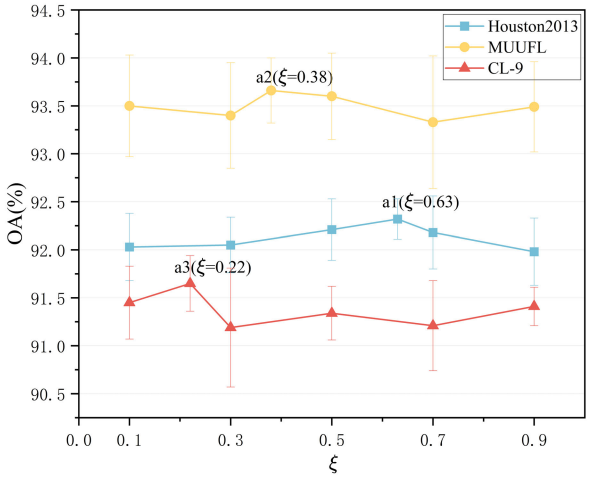


Fig. 14. Effects of ξ on the classification performance for three data.

robust stability. On the Houston2013 and CL-9 datasets, optimal model performance was observed near $\mu = 1.5$; when μ was either excessively large or small, model performance declined. For the MUUFL dataset, peak performance occurred at $\mu = 1.2$, which was marginally higher (by 0.02%) than the suboptimal setting of $\mu = 1.7$. Consequently, in our experiments, we selected $\mu = 1.5$ for the Houston2013 and CL-9 datasets and $\mu = 1.2$ for the MUUFL dataset.

3) Influence Analysis of the Fractional Order ξ: In the WFrFEF module, the WFrFT can transform HSI features into the fractional domain. In this domain, HSI features can more efficiently suppress noise and reduce information redundancy, thereby preserving more representative discriminative information for enhanced fusion with LiDAR features in subsequent stages. However, the fractional order parameter ξ, a crucial factor regulating the state of the fractional domain, may affect the model's classification performance across various datasets.

To this end, experiments were designed wherein the ξ value was selected from the set $\{0.1, 0.3, \ldots, 0.9\}$, along with an adaptive strategy for selecting the ξ value (a1, a2, and a3 represent the orders of adaptive learning for the datasets Houston2013, MUUFL and CL-9, respectively). Fig. 14 illustrates the trend of OA across three distinct datasets. The figure displays that when the adaptive selection of ξ is utilized, FrFENet achieves optimal classification performance on all three datasets. Conversely, when the ξ value is manually set, the Houston2013 and MUUFL datasets exhibit superior performance at $\xi \approx 0.5$, while the CL-9 dataset performs optimally at $\xi \approx 0.1$. This phenomenon further corroborates the unique requirements of different datasets for fractional domain characteristics. Consequently, the adaptive learning strategy for the ξ value aligns more closely with practical application scenarios, enabling the model to adapt flexibly to the characteristics of different datasets and achieve optimal classification performance.

F. Model Complexity Analysis

To systematically evaluate the model's performance, this section conducts assessments from three perspectives: spatial complexity, computational complexity, and temporal complexity. The evaluation is based on three key metrics: parameter quantities, floating-point operations per second (FLOPs), and time cost. As illustrated in the bubble chart (Fig. 15) and the starfish plot (Fig. 16), the complexity of various deep models is analyzed across three datasets. Experimental results indicate that CoCNN requires only 0.11, 0.1, and 0.1 M parameters on

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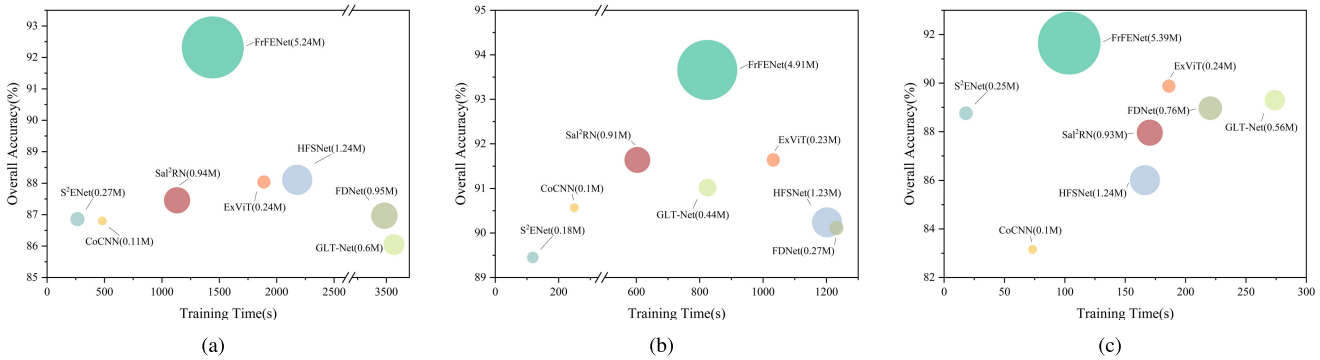


Fig. 15. Comparison of OA, parameter quantities, and average training time across three datasets. (a) Houston2013. (b) MUUFL. (c) CL-9.

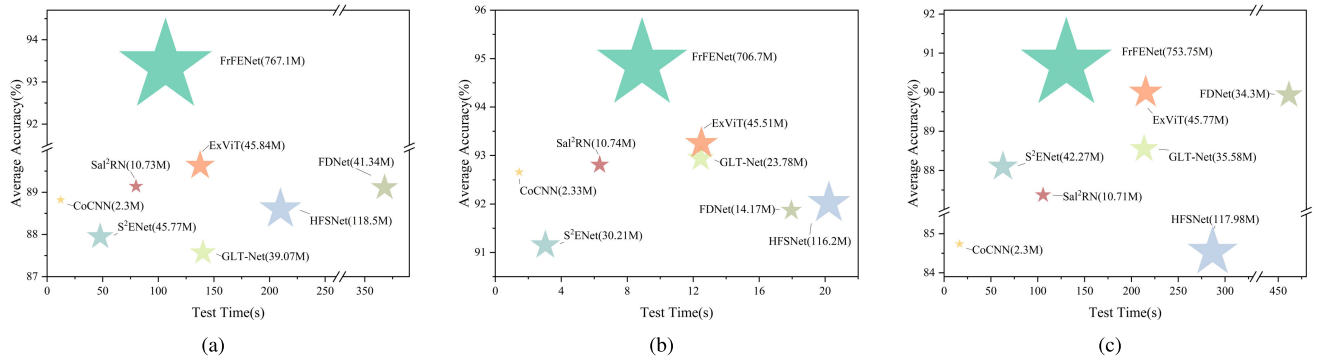


Fig. 16. Comparison of AA, FLOPs, and average test time across three datasets. (a) Houston2013. (b) MUUFL. (c) CL-9.

the Houston2013, MUUFL, and CL-9 datasets, respectively, making it the most parameter-efficient model (Fig. 15). This efficiency can be attributed to its coupled shared CNN layers and the absence of deep reasoning and interaction mechanisms, which also limit its accuracy. In contrast, FrFENet exhibits the highest parameter count, reaching 5.24, 4.91, and 5.39 M, due to its stacked ViT architecture, interaction mechanisms, and independent dual-branch structure. Notably, despite its large parameter count, FrFENet achieves a shorter model fitting time than comparison models ExViT/HFSNet/FDNet. The FLOPs comparison in Fig. 16 further validates this observation: CoCNN has the lowest FLOPs, whereas FrFENet has the highest. Nevertheless, our method achieves the shortest inference time during the testing phase compared to baseline models ExViT/HFSNet/FDNet while attaining the highest AA. In summary, given certain computational resources, FrFENet effectively optimizes the tradeoff between accuracy and efficiency, significantly enhancing classification performance in HSIC tasks.

IV. Conclusion

This article proposed the FrFENet based on Pareto optimization for the classification of HSI and LiDAR data. FrFENet can effectively mitigate the information swamping problem caused by modal information imbalance during the fusion stage. Additionally, FrFENet also effectively resists pixel disturbances within patches and alleviates the issue of under-optimization of one modality's model during joint learning of HSIs and LiDAR. Specifically, to mitigate the significant disparity in information content between different modalities, a WFrFEF is designed. This module enhances HSI feature representation and combines them with LiDAR features to generate balanced fused features. Additionally, to balance the learning rates of HSI and LiDAR features in a dual-branch neural network, a Pareto soft optimization strategy tailored for joint HSI and LiDAR classification, termed HLPareto, is introduced. To effectively utilize patch features that share the same land-cover label with the central pixel, an SSIM and an EIEM are developed during the HSI and LiDAR feature extraction stage. The SSIM facilitates the fusion of spatial–spectral features through token-level information interaction, effectively reducing the interference from irrelevant pixels. Meanwhile, the EIEM enhances the representation capability of elevation information while preserving the rich spatial geometric features in LiDAR data. Extensive experimental results indicate that FrFENet achieves the optimal classification performance among eight comparative methods, and ablation studies show the effectiveness of its different components.

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